Threshold Behaviour of a Three-Compartment Epidemic Model

A. Outbreak

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Abstract

We study the classical three-compartment epidemic in fractions: susceptible S, infectious I, and removed R. The total population is conserved, the outbreak grows only above the threshold $\beta/\gamma > 1$, and the infectious fraction peaks when the susceptible pool has crossed below the threshold value.

1 The model

With contact rate β and removal rate γ the dynamics read

$$\dot{S} = -\beta S I, \tag{1}$$

$$\dot{I} = \beta S I - \gamma I, \tag{2}$$

$$\dot{R} = \gamma I. \tag{3}$$

2 Initial state and parameters

The cohort starts almost fully susceptible, with the field-recorded initial state $S(0) = 0.99$, $I(0) = 0.01$ and $R(0) = 0$. The nominal parameters are $\beta = 0.4$ and $\gamma = 0.1$; at these values the threshold quantity β/γ exceeds one and an outbreak follows.

3 Discussion

Because removal never returns to the susceptible pool, the sum $S + I + R$ stays constant along every trajectory; the outbreak burns out by depleting the susceptible pool, leaving a final susceptible fraction well below its initial value.